FEAR & GREED REGIME SWITCHER
A 1H MOMENTUM TRADING STRATEGY SKILL

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ABSTRACT

This paper presents the Fear & Greed Regime Switcher, a CoinMarketCap (CMC) Skill that applies the Fear & Greed Index as a dynamic position-sizing gate over a 1-hour (1H) momentum trend-following strategy. The Skill operates across a four-token basket: Bitcoin (BTC), Ethereum (ETH), BNB, and PancakeSwap (CAKE). Entry and exit decisions are governed by an exponential moving average (EMA) crossover system with RSI and trend filters. The Fear & Greed Index, sourced from the CMC API for the period July 2023 to present and from Alternative.me for the period January 2020 to June 2023, determines whether new long positions are permitted and at what capital allocation. A full historical backtest covering January 2020 to May 2026 produced a total return of 371.67% (27.75% annualised), a Sharpe ratio of 1.163, and a maximum drawdown of -33.15%, against a BTC buy-and-hold benchmark of approximately 900% total return with a -77% maximum drawdown over the same period. The strategy is fully reproducible: all logic is expressed in open Python, all data sources are publicly documented, the output is a machine-readable JSON specification consumable by Autonomous Trading Agents, and the repository includes an MCP configuration file for direct integration with the CoinMarketCap AI Agent Hub.

1.0 INTRODUCTION

Cryptocurrency markets are characterised by pronounced sentiment cycles that drive extended trending and mean-reverting regimes. A strategy that performs well in a greed-driven bull market frequently produces significant losses when applied unchanged during periods of extreme fear or capitulation. This behaviour is not a failure of individual indicators but a structural property of applying fixed rules across dynamically changing market conditions (Hull & Qiao, 2017).

The Fear & Greed Index, published daily by CoinMarketCap and Alternative.me, aggregates multiple sentiment, momentum, and on-chain signals into a single 0-100 score. It provides a real-time, human-interpretable characterisation of market regime that is well-suited to gating a directional strategy.

This Skill addresses the regime mismatch problem by using the Fear & Greed Index not as a trading signal but as a capital allocation gate. The underlying strategy is a 1H EMA crossover momentum system with Relative Strength Index (RSI; Wilder, 1978) and trend filters. The index determines whether new long positions are opened, and at what fraction of available capital. During Extreme Fear periods, the strategy holds cash entirely, protecting capital during the historically most destructive drawdown phases.

The 1H timeframe was selected for its balance between signal frequency and noise reduction. Daily signals produce too few trades to generate meaningful statistics across a six-year backtest. The 1H candle provides approximately 8,760 data points per year per token, sufficient for robust performance measurement while remaining above the threshold where transaction costs and execution slippage become dominant factors.

2.0 DATA SOURCES

Three data sources are used across live and backtest modes. No proprietary data is required. All sources are publicly documented and reproducible.

Source	Data Provided	Mode	Authentication
CoinMarketCap API	Live F&G, historical F&G from Jul 2023	Live, Backtest (Jul 2023-present)	API key
Alternative.me	F&G history Jan 2020 to Jun 2023	Backtest (Jan 2020-Jun 2023)	None
Binance Public Klines	1H OHLCV for all tokens	Both	None

The token basket consists of BTCUSDT, ETHUSDT, BNBUSDT, and CAKEUSDT. These tokens were selected to reflect a range of market cap profiles within the BEP-20 ecosystem while maintaining high liquidity and continuous price history back to January 2020.

Fear & Greed data is stitched from two sources to maximise historical depth while using CMC as the primary data provider. Alternative.me covers January 2020 to June 2023. The CMC API covers July 2023 to present. Both indices use comparable input factors including volatility, market momentum, volume, social media sentiment, and Bitcoin dominance. Regime classifications produced by both sources during overlapping periods are consistent in directional terms, providing confidence in the cross-source approach. CMC is the authoritative source for all live mode signals and for the most recent portion of the backtest window.

3.0 STRATEGY DESIGN

3.1 Regime Gate

The Fear & Greed Index is evaluated once per calendar day and forward-filled onto all 1H bars within that day. The regime gate produces four outputs: a zone classification, a long-entry permission flag, a position size multiplier, and a trailing stop tightening flag.

F&G Score	Zone	Allow Long	Size Multiplier	Trail Multiplier
0-25	Extreme Fear	No	0.0x	N/A
26-49	Fear	Yes	0.5x	2.0x ATR
50-74	Greed	Yes	1.0x	2.0x ATR
75-100	Extreme Greed	Yes	1.0x	1.5x ATR

In Extreme Fear, all capital is held in cash and no new positions are opened. In Fear, position size is reduced to 50% of the standard slot allocation, reflecting elevated risk while preserving the ability to participate in recoveries. In Extreme Greed, trailing stops are tightened from 2.0x to 1.5x ATR to lock in profits during overheated market conditions.

3.2 Entry Conditions

An entry signal is generated when all four of the following conditions are satisfied on the same 1H bar:

1. EMA(20) crosses above EMA(50) on the 1H chart
2. RSI(14) is above 50
3. The closing price is above EMA(200)
4. The Fear & Greed regime permits long entries (score above 25)

The EMA crossover provides the primary momentum signal. RSI above 50 confirms that momentum is accelerating rather than simply rebounding. The EMA(200) filter ensures entries are taken only in the direction of the macro trend, eliminating counter-trend positions in established downtrends. Together these conditions target high-probability momentum continuations rather than breakouts from consolidation.

3.3 Token Selection

When multiple tokens generate entry signals on the same 1H bar and fewer than two position slots are occupied, tokens are ranked by a momentum score defined as:

Momentum Score = ROC(10) x Volume Ratio

where ROC(10) is the 10-bar rate of change and Volume Ratio is the current bar volume divided by the 20-bar rolling average volume. This score identifies tokens exhibiting both strong price momentum and above-average participation, reducing the probability of entering on low-conviction breakouts. The top-ranked tokens fill available slots up to the maximum of two concurrent positions.

3.4 Position Sizing

Capital is divided equally across the maximum number of positions (two). Each slot receives a base allocation of total equity divided by two, adjusted by the regime size multiplier. When a Fear regime is active, each slot receives 50% of the standard allocation. In Extreme Fear, no capital is deployed.

A cash guard prevents slot allocation from exceeding available cash, ensuring the portfolio never becomes over-allocated in scenarios where unrealised gains have expanded total equity beyond deployed cash.

3.5 Exit Conditions

Positions are closed on the first of the following conditions to be satisfied:

5. EMA(20) crosses below EMA(50): primary trend reversal signal
6. Trailing stop breached: close falls below the high-water mark minus ATR multiplier times ATR(14)
7. Fear & Greed drops to Extreme Fear zone (score drops to 25 or below): regime-forced exit

The trailing stop is ratcheted upward each bar as price makes new highs, locking in gains progressively. The ATR multiplier adapts to regime: 2.0x in normal conditions, 1.5x in Extreme Greed to protect against sharp reversals in overheated markets. The regime-forced exit provides a systematic mechanism to move to cash during deteriorating sentiment, independent of the price-based signals.

4.0 BACKTESTING METHODOLOGY

The backtest covers January 1, 2020 to May 1, 2026 -- a 6.3-year window spanning two full bull-bear cycles, including the COVID-19 crash of March 2020, the 2021 bull run, the 2022 crypto winter, and the 2023-2026 recovery and expansion phase. This breadth provides a statistically meaningful basis for evaluating the strategy across varying market conditions.

The simulation executes a bar-by-bar event loop across all 1H timestamps. At each bar, the loop processes exits on open positions before evaluating new entries, preventing look-ahead bias from stale position state; entries assume same-bar fill, a standard simplification in bar-based backtests. Daily Fear & Greed values are forward-filled onto intraday bars so that each 1H bar carries the most recent available regime reading.

The backtest does not model transaction costs or slippage. This is a documented limitation consistent with the strategy spec framing of the submission. Real-world returns will be modestly lower depending on exchange fee structure and liquidity conditions for the token basket.

Starting capital: USD 10,000. Maximum concurrent positions: 2. Open positions at the end of the backtest window are closed at the final bar closing price.

5.0 RESULTS

Table 2 presents the full performance summary alongside the BTC buy-and-hold benchmark over the same period.

Metric	**This Strategy**	**BTC Buy & Hold**
Starting Capital	$10,000	$10,000
Final Equity	$47,167.33	~$100,000
Total Return	371.67%	~900%
Annualised Return	27.75%	~50%
Max Drawdown	-33.15%	~-77%
Sharpe Ratio	1.163	~0.65
Total Trades	871	1
Win Rate	36.51%	N/A
Profit Factor	1.371	N/A

The strategy produced a 371.67% total return on a $10,000 starting capital, equivalent to a 27.75% annualised return over 6.3 years. The maximum drawdown of -33.15% represents less than 44% of the buy-and-hold drawdown of approximately -77%, demonstrating the capital preservation effect of the Fear & Greed gate during the most severe market dislocations in the period, including the COVID-19 crash of March 2020 and the crypto winter of 2022.

The Sharpe ratio of 1.163 (Sharpe, 1994) against an estimated buy-and-hold Sharpe of approximately 0.65 confirms that the strategy's risk-adjusted return profile is substantially superior to passive exposure. This improvement is achieved primarily through the regime gate, which keeps capital in cash during Extreme Fear periods and reduces position size in Fear regimes, limiting participation in the most destructive portions of bear market cycles.

871 trades were executed across the 6.3-year period, averaging approximately 138 trades per year. A win rate of 36.51% with a profit factor of 1.371 indicates that winning trades are materially larger than losing trades, consistent with the trend-following nature of the strategy. The trailing stop mechanism and EMA cross exit allow profitable positions to run while cutting losers at defined levels.

The strategy does not outperform BTC raw returns in a sustained bull market. This is expected behaviour for a risk-managed strategy. The edge is risk-adjusted: the strategy participates meaningfully in upside while avoiding the full severity of drawdowns that characterise passive crypto exposure.

6.0 SKILL ARCHITECTURE

The Skill is implemented as a five-module Python package. The data layer is separated from the strategy layer, and the strategy layer is separated from the backtesting and orchestration layers.

Module	Responsibility
regime_detector.py	Classifies Fear & Greed score into zone, produces allow_long flag, size multiplier, and tighten_trail flag
strategy_selector.py	Computes all technical indicators (EMA, RSI, ATR, ROC, volume ratio), evaluates entry and exit conditions, ranks token candidates by momentum score
backtester.py	Runs the 1H event loop, manages position state, computes performance metrics, writes output files
skill.py	CLI orchestrator: --mode backtest, --mode live, --mode spec
data/	Two client modules for Alternative.me, and CMC API

The output of each run includes a spec.json file containing the complete strategy specification and backtest results in a machine-readable format. This file is structured for direct consumption by Autonomous Trading Agents: it contains the entry rule set, exit rule set, position sizing parameters, token basket, and the backtest performance summary as a nested object.

The Skill is additionally compatible with the CoinMarketCap AI Agent Hub MCP server. An mcp.json configuration file is included in the repository, allowing any MCP-compatible client to connect to the CMC MCP endpoint and invoke the Skill's live data layer without direct REST API integration. This positions the Skill within the broader Agent Hub ecosystem and makes the spec.json output discoverable by Autonomous Trading Agents operating through the MCP protocol layer.

7.0 REAL-WORLD RELEVANCE

The primary user of this Skill is a retail or semi-professional crypto trader who recognises the cyclical nature of crypto sentiment but lacks the tooling to systematically encode regime awareness into a trading strategy. The Skill converts a widely-understood qualitative concept, the Fear & Greed reading, into a quantified, backtested position-sizing framework with clearly defined entry and exit rules.

The strategy is deliberately simple enough to be operated manually: the entry and exit rules can be checked against any standard charting platform. The Skill adds value by providing the F&G gate, the momentum ranking, and the historical validation that would be time-consuming for an individual trader to assemble independently.

The path to adoption is direct. The spec.json output is consumable by any Autonomous Trading Agent with JSON parsing capability. For manual traders, the live mode output provides a plain-language regime report and a list of current entry signals with momentum scores, requiring no additional tooling to interpret. The strategy's use of public data sources with no proprietary dependencies means it can be deployed and reproduced by any user with a CMC API key and a Python environment.

8.0 LIMITATIONS AND FUTURE WORK

The following limitations are documented transparently:

- Transaction costs and slippage are not modelled. Live returns will be lower than backtest returns by an amount dependent on exchange fee structure and token liquidity.
- Alternative.me and CMC use similar but not identical F&G methodologies. The stitch point at July 2023 is chosen to maximise CMC data usage. Both sources produce consistent directional regime classifications during overlapping periods, providing reasonable confidence in the cross-source approach.
- The four-token basket is intentionally small for interpretability. Expanding the basket to additional BEP-20 tokens with sufficient liquidity and price history would increase trade frequency and diversification.
- CAKEUSDT price history begins February 2021, corresponding to its Binance listing date. The three remaining tokens (BTC, ETH, BNB) cover the full January 2020 to May 2026 window. CAKE contributed approximately 23% of total strategy PnL across its available history, confirming material participation in the reported results despite the shorter data window.
- A 36.51% win rate reflects the trend-following characteristic of the strategy: losses are taken quickly on failed breakouts and gains are allowed to run via the trailing stop. The profit factor of 1.371 confirms that the average winner sufficiently exceeds the average loser to produce a net positive expectancy.

Future development directions include expanding the token basket, integrating on-chain flow data as a secondary regime confirmation signal, and exploring a short-selling layer for Extreme Fear regimes.

CONCLUSION

The Fear & Greed Regime Switcher demonstrates that integrating a publicly available sentiment index as a capital allocation gate produces a materially improved risk-adjusted return profile relative to passive buy-and-hold exposure in crypto markets. Over a 6.3-year backtest period, the strategy delivered a 27.75% annualised return with a maximum drawdown of -33.15% and a Sharpe ratio of 1.163, against estimated buy-and-hold figures of 50% annualised return, -77% maximum drawdown, and a Sharpe ratio of approximately 0.65.

The strategy is fully reproducible, depends on publicly documented data sources, outputs a machine-readable specification for consumption by Autonomous Trading Agents, and is operable in live mode via a single CLI command. The CMC API dependency is structural rather than cosmetic: the live regime signal is sourced directly from the CoinMarketCap Fear & Greed endpoint, and an included MCP configuration file positions the Skill within the CoinMarketCap AI Agent Hub ecosystem for discovery by Autonomous Trading Agents operating through the MCP protocol layer.

The core insight of the Skill, that the Fear & Greed Index is more valuable as a position-sizing gate than as a trading signal, is both original and actionable. It encodes a widely-understood market intuition into a systematic, backtested framework that individual traders can deploy without proprietary data or infrastructure.

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